

2/11/2015.

51.504 Maximum Finding

$$1) f(x, y) = y - 3x$$

$$\text{constraint} \equiv x^2 + y^2 = 10$$

$$(a) L(x, y, \lambda) = y - 3x + \lambda(x^2 + y^2 - 10)$$

$$(b) \frac{\partial L}{\partial x} = 0, \frac{\partial L}{\partial y} = 0, x^2 + y^2 = 10 \Rightarrow \text{for extremum.}$$

$$(i) -3 + \lambda(2x) = 0$$

$$\Rightarrow \lambda(2x) = 3$$

$$\Rightarrow x = \frac{3}{2\lambda}$$

$$(ii) 1 + \lambda(2y) = 0$$

$$\Rightarrow 1 + 2\lambda y = 0$$

$$\Rightarrow y = -\frac{1}{2\lambda}$$

$$\Rightarrow 10 = 10 - 4\lambda^2$$

$$\Rightarrow 10(1 - 4\lambda^2) = 0$$

$$\Rightarrow 1 - 4\lambda^2 = 0$$

$$\Rightarrow 4\lambda^2 = 1$$

$$\Rightarrow \lambda = \pm \frac{1}{2}$$

$$\text{Therefore solutions} = \overset{\text{I}}{(-3, 1)}, \overset{\text{II}}{(3, -1)}$$

$$f(\text{I}) = 1 - 3(-3) = 10$$

$$f(\text{II}) = -1 - 3(3) = -10 \rightarrow \text{minimum}$$

$$\therefore \text{Min occurs @ } (3, -1)$$

(c) for inequality condition we can impose $\lambda \geq 0$.

The basic form of the Lagrangian remains the same.

$$L(x, y, \lambda) = y - 3x + \lambda(x^2 + y^2 - 10)$$

$$\boxed{\lambda \geq 0}$$

(d) KKT / complementary slackness conditions:

(i) $-2 + 2\lambda x = 0$, $1 + 2\lambda y = 0$ [stationary]

(ii) Primary constraint: $x^2 + y^2 = 10$

(iii) $\lambda \geq 0$

(iv) $\lambda(x^2 + y^2 - 10) = 0$

If the constraint is active ($x^2 + y^2 = 10$) λ is positive

OR

If constraint is inactive ($x^2 + y^2 < 10$)

(2) (a) Linear separator equation: $\text{signum}(\theta_1 x + \theta_0)$
 $\therefore \theta_1 = 2, \theta_0 = -3 \Rightarrow \text{signum}(2x - 3)$

(b) x is input, y is output.

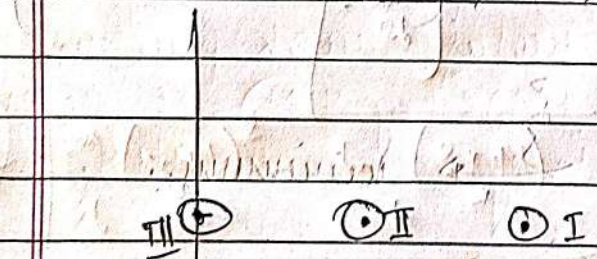
(I) $(2, 1) \rightarrow 1$

(II) $(1, 0) \rightarrow -1$

(III) $(0, 1) \rightarrow -1$

(IV) $(-2, -1) \rightarrow -1$

} (misclassified)



(IV)

$$(c) \quad \theta_1' = \theta_1 + \alpha y^{(1)} x^{(1)} \\ \theta_0' = \theta_0 + \alpha y^{(1)}$$

$$\text{choose } (1, 1) \longrightarrow \begin{cases} \theta_1' = \theta_1 + \alpha = 2 + \alpha \\ \theta_0' = \theta_0 + \alpha = -3 + \alpha \end{cases}$$

$$\text{choose } (0, 1) \longrightarrow \begin{cases} \theta_1' = 2 + 0 = 2 \\ \theta_0' = -3 + \alpha = -3 + \alpha \end{cases}$$

$$(d) \text{ Use same equation } \longrightarrow \theta_1 x + \theta_0 \Rightarrow \theta_1' x + \theta_0'$$

$$\therefore \text{ choosing } (1, 1) \longrightarrow (2 + \alpha)x + (-3 + \alpha) = 0$$

$$u(1, 1) > 1 \Rightarrow (2 + \alpha) + (-3 + \alpha) = -1 + 2\alpha > 0$$

$$2\alpha > 1 \Rightarrow \alpha > \frac{1}{2}$$

$$u(0, 1) > 1 \Rightarrow (2 + \alpha) + (-3 + \alpha) > 0$$

$$\Rightarrow -3 + \alpha > 0 \Rightarrow \boxed{\alpha > 3}$$

$\therefore \forall \alpha > 3$, for the perceptron updation algorithm we will have correct results. (ONE step correct)

$$\text{choosing } (0, 1) \longrightarrow 2x + (-3 + \alpha)$$

$$\begin{aligned} u(1, 1) = 2 - 3 + \alpha > 0 &\Rightarrow \alpha > 1 \\ u(0, 1) = -3 + \alpha > 0 &\Rightarrow \alpha > 3 \end{aligned} \quad \left. \vphantom{\begin{aligned} u(1, 1) = 2 - 3 + \alpha > 0 \\ u(0, 1) = -3 + \alpha > 0 \end{aligned}} \right\} \boxed{\alpha > 3}$$

$\therefore$ irrespective of which perceptron algorithm we choose the condition for $\boxed{\alpha > 3}$ for $\forall \alpha \in \mathbb{R}$.

Date / /

$$(3) \quad K(x, y) = (xy + 1)^2$$

$$\begin{aligned} x^{(1)}, t^{(1)} &= (1, 1) \\ x^{(2)}, t^{(2)} &= (3, -1) \end{aligned} \rightarrow \text{sampled pts.} \quad \left(x = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, t = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right)$$

Let, test input $\Rightarrow x^*$

$$Cov(K) = C = \begin{bmatrix} K(1,1) & K(1,3) \\ K(3,1) & K(3,3) \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 16 \\ 16 & 100 \end{bmatrix}$$

$$\text{Mean: } \mu(x) = \frac{1}{N} C^{-1} t$$

$$= \frac{1}{N} \begin{bmatrix} (x+1)^2 & (3x+1)^2 \end{bmatrix} C^{-1} t$$

$$= \frac{1}{N} \begin{bmatrix} (x+1)^2 & (3x+1)^2 \end{bmatrix} \begin{bmatrix} 25/36 & -1/9 \\ -1/9 & 1/36 \end{bmatrix} t$$

$$\det(C) = 400 - 256$$

$$C^{-1} = \frac{1}{144} \begin{bmatrix} 100 & -16 \\ -16 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 25/36 & -1/9 \\ -1/9 & 1/36 \end{bmatrix}$$

$$C^{-1} t = \begin{bmatrix} 25/36 & -1/9 \\ -1/9 & 1/36 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 25/36 + 1/9 \\ -1/9 + 1/36 \end{bmatrix} = \begin{bmatrix} 29/36 \\ -1/36 \end{bmatrix}$$

$$= \begin{bmatrix} 25/36 & -1/9 \\ -1/9 & 1/36 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 25/36 + 1/9 \\ -1/9 + 1/36 \end{bmatrix} = \begin{bmatrix} 29/36 \\ -1/36 \end{bmatrix}$$

$$\mu(x) = \frac{29}{36} (x+1)^2 - \frac{1}{36} (3x+1)^2$$

6

(5)

saathi

Date ___/___/___

$$\Delta^2 = K(x, x) - K^T C^{-1} K$$

$$K = \begin{bmatrix} (x+1)^2 \\ (3x+1)^2 \end{bmatrix}$$

$$C^{-1} = \begin{bmatrix} \frac{25}{36} & -\frac{1}{9} \\ -\frac{1}{9} & \frac{1}{36} \end{bmatrix}$$

$$\therefore C^{-1} K = \begin{bmatrix} \frac{25}{36} (x+1)^2 - \frac{1}{9} (3x+1)^2 \\ -\frac{1}{9} (x+1)^2 + \frac{1}{36} (3x+1)^2 \end{bmatrix}$$

$$K^T \begin{bmatrix} (x+1)^2 & (3x+1)^2 \end{bmatrix} \sim [a \quad b]$$

$$\text{Quinary } C^{-1} K = \begin{bmatrix} \frac{25}{36} a - \frac{1}{9} b \\ -\frac{1}{9} a + \frac{1}{36} b \end{bmatrix}$$

$$K^T C^{-1} K = [a \quad b] \begin{bmatrix} \frac{25}{36} a - \frac{1}{9} b \\ -\frac{1}{9} a + \frac{1}{36} b \end{bmatrix} = a \left(\frac{25}{36} a - \frac{1}{9} b \right) + b \left(-\frac{1}{9} a + \frac{1}{36} b \right)$$

$$K(x, x) = (x \cdot x + 1)^2 = (x^2 + 1)^2$$

$$K^T C^{-1} K = \frac{25}{36} a^2 - \frac{ab}{9} - \frac{ab}{9} + \frac{b^2}{36} = \frac{25a^2 - 8ab + b^2}{36}$$

$$\Delta^2 = (x^2 + 1)^2 - \frac{1}{36} [25(x^2 + 1)^4 - 8(x+1)^2(3x+1)^2 + (3x+1)^4]$$

(b) Kernel function MUST be positive semidefinite. which means for ANY kernel K , $K^T K \geq 0$. (equality for "semi" definite, greater for strictly definite)
 Now for $K = (xy - 5) \rightarrow K(0,0) = 0 \cdot 0 - 5 = -5 < 0$.
 Hence, $xy - 5$ is INVALID kernel.